Domain and Range

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domain from equations**

State the domain of each function:

a $f(x) = \frac{1}{x-4}$ All real x with $x \neq 4$.

b $g(x) = \sqrt{2x-6}$ $2x - 6 \geq 0$, so $x \geq 3$.

c $h(x) = \frac{x+1}{x^2-9}$ All real x with $x \neq 3$ and $x \neq -3$.

d $k(x) = \sqrt[3]{x}$ All real numbers.

(a) All real x with $x \neq 4$.; (b) $2x - 6 \geq 0$, so $x \geq 3$.; (c) All real x with $x \neq 3$ and $x \neq -3$.; (d) All real numbers.

2 **[Foundational]** • **Target: Domain from equations**

Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

We need $5 - x \geq 0$, so $x \leq 5$, and $x \neq 1$. Domain: $(-\infty, 1) \cup (1, 5]$.

3 **[Foundational]** • **Target: Domain and range from graphs**

The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.

Domain: all real numbers. Range: $y \geq -2$.

4 **[Foundational]** • **Target: Finding the range algebraically**

By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

$f(x) = (x + 2)^2 - 3$. Since $(x + 2)^2 \geq 0$, the range is $y \geq -3$.

5 **[Foundational]** • **Target: Finding the range algebraically**

Consider $f(x) = -2(x - 1)^2 + 5$.

a State the coordinates of the vertex. Vertex: $(1, 5)$.

b Does the parabola open upward or downward? Downward, since the leading coefficient $-2 < 0$.

c State the range of f. $y \leq 5$.

(a) Vertex: $(1, 5)$.; (b) Downward, since the leading coefficient $-2 < 0$.; (c) $y \leq 5$.

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6 [Foundational] • Target: Finding the range algebraically

Find the range of $g(x) = \sqrt{9 - x^2}$ (the upper half of a circle).

The domain is $[-3, 3]$; g increases from $g(\pm 3) = 0$ to a maximum $g(0) = 3$. Range: $0 \le y \le 3$.

7 [Foundational] • Target: Domain and range in applied contexts

A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20 - x)$.

a State the domain of A in this context (both dimensions must be positive). $0 < x < 20$.

b Find $A(6)$ and $A(15)$. $A(6) = 6(14) = 84$ m^2; $A(15) = 15(5) = 75$ m^2.

c State the range of A over this domain, to the nearest whole number. $A(x) = -(x - 10)^2 + 100$ has maximum 100 at $x = 10$ and approaches 0 at the endpoints: range $0 < y \le 100$.

(a) $0 < x < 20$.; (b) $A(6) = 6(14) = 84$ m^2; $A(15) = 15(5) = 75$ m^2.; (c) $A(x) = -(x - 10)^2 + 100$ has maximum 100 at $x = 10$ and approaches 0 at the endpoints: range $0 < y \le 100$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range in applied contexts

An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12 - 2x)^2$.

a State the domain of V in this physical context. $0 < x < 6$.

b Find $V(1)$ and $V(4)$. $V(1) = 1(10)^2 = 100$ cm^3; $V(4) = 4(4)^2 = 64$ cm^3.

(a) $0 < x < 6$.; (b) $V(1) = 1(10)^2 = 100$ cm^3; $V(4) = 4(4)^2 = 64$ cm^3.

9 [Intermediate] • Target: Domain of composite and combined functions

Find the domain of $f(x) = \frac{1}{\sqrt{x^2 - 4}}$, writing your answer in interval notation.

Need $x^2 - 4 > 0$ (strictly, since it's in a denominator under a root): $x < -2$ or $x > 2$. Domain: $(-\infty, -2) \cup (2, \infty)$.

10 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $f(x) = \frac{x^2 - 4}{x^2 - x - 6}$, factoring the denominator first.

$x^2 - x - 6 = (x - 3)(x + 2)$. Domain: all real x with $x \neq 3$ and $x \neq -2$.

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11 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $g(x) = \frac{\sqrt{x+4}}{x-1}$, combining a root restriction with a denominator restriction.

Need $x + 4 \geq 0 \Rightarrow x \geq -4$, and $x \neq 1$. Domain: $[-4, 1) \cup (1, \infty)$.

12 [Intermediate] • Target: Range using transformations

The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.

$g(x) = x^2 - 5$ shifts the range down 5: range $y \geq -5$. $h(x) = -x^2$ reflects the range over $y = 0$: range $y \leq 0$.

13 [Intermediate] • Target: Range using transformations

The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.

Multiplying by 3 scales the range but preserves $y \geq 0$; adding 2 shifts up: range $y \geq 2$.

14 [Intermediate] • Target: Domain and range of piecewise functions

A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.

Domain: all real numbers (every x has a defined piece). For $x < 0$: $f(x) > 0$ (excluding 0 itself, approaching it). For $x \geq 0$: $f(x) \geq 1$. Combined range: $y > 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Range using transformations • Cross-Lesson: 5.03 Parametric Equations

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x-2)(x+2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3.$ The horizontal asymptote is $y = 3$. Notice tha.

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16 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

17 [Challenging] • Target: Domain from equations • Cross-Lesson: 6.02 Polar Equations and Graphs

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Set up the decomposition: $\frac{7x - 1}{(x - 3)(x + 2)} = \frac{A}{x - 3} + \frac{B}{x + 2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x - 1}{x + 2}\Big|_{x = 3} = \frac{7(3) - 1}{3 + 2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x - 1}{x - 3}\Big|_{x = -2} = \frac{7(-2) - 1}{-2 - 3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x - 3} + \frac{3}{x + 2}$.

18 [Challenging] • Target: Domain and range from graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Finding the range algebraically • Cross-Lesson: 7.02 Logarithms and Their Properties

Level 3 · AdvancedLet $R(x) = \frac{2x^2 - 8}{x^2 - 5x + 6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

(a) Domain: Factor the denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Thus the domain is all real numbers except $x \neq 2$ and $x \neq 3$, or $(-\infty, 2) \cup (2, 3) \cup (3, \infty)$. (b) Holes: Factor the numerator: $2(x^2 - 4) = 2(x - 2)(x + 2)$. The factor $(x - 2)$ cancels: $R(x) = \frac{2(x + 2)}{x - 3} \quad (x \neq 2)$. Evaluating at $x = 2$: $y = \frac{2(2 + 2)}{2 - 3} = \frac{8}{-1} = -8$. Hole coordinates: $(2, -8)$. (c) Asymptotes: Vertical asymptote at the remaining denominator zero: $x = 3$. Degrees of nu.

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20 [Challenging] • Target: Domain and range in applied contexts • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{9^2+10^2-7^2}{2(9)(10)} = \frac{81+100-49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7+9+10}{2} = 13$..

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain from equations

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. **27** ✓

d. 26

By the Remainder Theorem, the remainder equals $f(-3) = 2(-3)^2 - 1(-3) + 6 = 27$.

22 [Foundational] • Target: Domain and range from graphs

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

a. $\pm 1, \pm 2$

b. ± 6

c. ± 12

d. $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$ ✓

Possible rational roots are $\pm\frac{p}{q}$ where p divides the constant term 6 and q divides the leading coefficient 2: $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$.

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23 [Foundational] • Target: Finding the range algebraically

Find all vertical asymptotes of $f(x) = \frac{x^2 + x - 6}{x^3 - 6x^2 + 3x + 10}$.

a. x=2,-1,5

b. x=2

c. x=-1

d. **x=-1,5 ✓**

Factor: $f(x) = \frac{(x-2)(x+3)}{(x-2)(x-5)(x+1)}$. The $(x-2)$ factor cancels (hole at $x = 2$), leaving vertical asymptotes at $x = -1, 5$.

24 [Foundational] • Target: Domain and range in applied contexts

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = -3$, vertical asymptote at $x = -1$

b. **Hole at $x = -1$, vertical asymptote at $x = -3$ ✓**

c. Vertical asymptotes at $x = -1$ and $x = -3$, no hole

d. Hole at $x = -2$, vertical asymptote at $x = -3$

The factor $(x + 1)$ cancels, creating a hole at $x = -1$; the remaining zero of the denominator, $x = -3$, gives the vertical asymptote.

25 [Foundational] • Target: Domain of composite and combined functions

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. **crosses the x-axis, flattening (S-shaped) near the zero ✓**

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

The zero $x = 4$ has multiplicity 3, which is odd, so the graph crosses the x-axis, flattening (S-shaped) near the zero.

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26 [Foundational] • Target: Domain from rational functions with factoring

As $x \to +\infty$, what happens to $f(x) = 2x^5$?

a. $+\infty$ ✓

b. $-\infty$

c. 0

d. Oscillates

Leading coefficient is 2 (positive), degree 5. As $x \to +\infty$, $f(x) \to +\infty$.

27 [Foundational] • Target: Range using transformations

Find the slant (oblique) asymptote of $f(x) = \frac{x^2+5x+1}{x-1}$.

a. $y = 6x$

b. $y = x + 6$ ✓

c. $y = x + 7$

d. No slant asymptote

Divide: $x^2 + 5x + 1 = (x-1)(x+6) + 7$. Dropping the remainder term, the slant asymptote is $y = x + 6$.

28 [Foundational] • Target: Domain and range of piecewise functions

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2+x-6}{x^2-7x+10}$.

a. x=5

b. **x=2** ✓

c. No hole exists

d. x=-3

$f(x) = \frac{(x-2)(x+3)}{(x-2)(x-5)}$; the common factor $(x-2)$ gives a hole at $x = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Domain from equations

Find the horizontal asymptote of $f(x) = \frac{x^3+1}{x^2-4}$, or state that none exists.

a. The line $y = x$

b. y=0

c. y=1

d. No horizontal asymptote ✓

Degree of numerator $>$ degree of denominator $\Rightarrow$ no horizontal asymptote.

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30 [Intermediate] • Target: Domain and range from graphs

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

31 [Intermediate] • Target: Finding the range algebraically

What is the horizontal asymptote of $f(x) = \frac{1}{x}$?

a. $y = 1$

b. $y = 0$ ✓

c. None

d. $x = 0$

Degree of numerator (0) is less than degree of denominator (1), so $y = 0$.

32 [Intermediate] • Target: Domain and range in applied contexts

What is the vertical asymptote of $f(x) = \frac{1}{x-0}$?

a. $x = 0$ ✓

b. $y = 0$

c. $x = 1$

d. None

The denominator is zero at $x = 0$, giving a vertical asymptote there.

33 [Intermediate] • Target: Domain of composite and combined functions

Find the horizontal asymptote of $f(x) = \frac{3}{x^2 - 4}$.

a. **y=0** ✓

b. y=-1

c. y=1

d. y=3

Degree of numerator $<$ degree of denominator $\Rightarrow y = 0$.

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34 [Intermediate] • Target: Domain from rational functions with factoring

Simplify $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$ and evaluate at $x = 1$.

a. 1

b. 0

c. 2

d. **-1 ✓**

$f(x) = \frac{x+3}{x-5}$ for $xeq2$; at $x = 1$: $\frac{4}{-4} = -1$.

35 [Intermediate] • Target: Range using transformations

What is the degree and leading coefficient of $f(x) = 2x^4 + x^2 - 2$?

a. Degree 5, leading coefficient 2

b. **Degree 4, leading coefficient 2 ✓**

c. Degree 4, leading coefficient -2

d. Degree 3, leading coefficient 2

The highest power of x is 4, with coefficient 2.

36 [Intermediate] • Target: Domain and range of piecewise functions

Find the vertical asymptote of $f(x) = \frac{2}{x-3}$.

a. y=3

b. x=0

c. **x=3 ✓**

d. x=-3

Denominator $= 0 \Rightarrow x - 3 = 0 \Rightarrow x = 3$.

37 [Intermediate] • Target: Domain from equations

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Domain and range from graphs • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

39 [Challenging] • Target: Finding the range algebraically • Cross-Lesson: 7.02 Logarithms and Their Properties

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$ ✓

$(45^+30^) = 45^30^+45^30^= 6 + 24$.

40 [Challenging] • Target: Domain and range in applied contexts • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$ ✓

b. $\frac{1}{2}$

c. 22

d. $6 - 24$

$(60^+45^) = 60^45^-60^45^= 2 - 64$.

41 [Challenging] • Target: Domain of composite and combined functions • Cross-Lesson: 7.04 Applications: Growth and Decay

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.

a. **5.7** ✓

b. 6.7

c. 7.7

d. 4.7

By the Law of Sines: $\frac{a}{\sin A} = \frac{b}{\sin B}$, so $b = \frac{5\sin 80^\circ}{\sin 60^\circ} \approx 5.7$.

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42 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 8.01 Rational Functions and Asymptotes

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD. ✓

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

Let $\angle BAD = \angle CAD = \alpha$. In triangle ABD, by the Sine Rule, we have $\frac{m}{\sin\alpha} = \frac{c}{\sin\angle ADB}$. Therefore, $m = \frac{c\sin\alpha}{\sin\angle ADB}$. In triangle ACD, by the Sine Rule, we have $\frac{n}{\sin\alpha} = \frac{b}{\sin\angle ADC}$. Therefore, $n = \frac{b\sin\alpha}{\sin\angle ADC}$. Since $\angle ADB$ and $\angle ADC$ are supplementary angles, $\sin\angle ADB = \sin(180^\circ - \angle ADC) = \sin\angle ADC$. Now, we can form the ratio.

43 [Challenging] • Target: Range using transformations • Cross-Lesson: 8.02 Graphing Rational Functions

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

a. Consistent and independent (one solution)

b. Consistent and dependent (infinitely many solutions) ✓

c. Inconsistent (no solution)

d. Cannot be determined

Different slopes give exactly one intersection (independent). Equal slopes with equal intercepts give the same line (dependent, infinite solutions). Equal slopes with different intercepts give parallel lines (inconsistent, no solution).

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44 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 8.03 Introduction to Limits

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$ ✓

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{10}{8}\langle 2, 2 \rangle = \left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$.

45 [Challenging] • Target: Domain from equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$ ✓

d. $\frac{20}{29}$

By the Pythagorean identity, $\cos\theta = \pm\sqrt{1 - \sin^2\theta} = \pm\frac{21}{29}$. In Quadrant 3, cosine is negative, so $\cos\theta = -\frac{21}{29}$.

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